

Supplementary Material

Step-by-step derivations for “Diffusion anisotropy and entropy: reflections of each other”

Scott N. Hwang, Jonathan K. Maffie, Sangam G. Kanekar

This document derives every relation stated in the paper, one step at a time, in the paper’s notation. Equation numbers without an S refer to the paper. Each derivation ends with the name of the check in `analysis/identity_checks.py` of the code repository that verifies it numerically.

S1 Setup

Let $x_1, \dots, x_K$ be positive numbers. Their mean is $\bar{x} = \frac{1}{K} \sum_i x_i$. The average of any quantity over i is written $\langle \cdot \rangle$, so $\bar{x} = \langle x \rangle$. The standard deviation uses divisor K ,

$$\sigma_x^2 = \frac{1}{K} \sum_i (x_i - \bar{x})^2 = \langle x^2 \rangle - \bar{x}^2, \quad (\text{S1})$$

and the coefficient of variation is $\text{CV}_x = \sigma_x / \bar{x}$. Dividing Eq. (S1) by $\bar{x}^2$ gives

$$\frac{\langle x^2 \rangle}{\bar{x}^2} = 1 + \text{CV}_x^2. \quad (\text{S2})$$

The proportions are $p_i = x_i / \sum_j x_j$. They are positive and sum to 1, so they form a probability distribution. When all values are equal, every p_i equals $1/K$. This is the uniform distribution u , with $u_i = 1/K$. Because $\sum_j x_j = K\bar{x}$,

$$p_i = \frac{x_i}{K\bar{x}}. \quad (\text{S3})$$

The Rényi entropy of order $\alpha \neq 1$ is $H_\alpha(p) = (1 - \alpha)^{-1} \ln \sum_i p_i^\alpha$. The Shannon entropy $H_1 = -\sum_i p_i \ln p_i$ is its limit as $\alpha \rightarrow 1$ (Section S4). The power mean of order $r \neq 0$ is $M_r(x) = \langle x^r \rangle^{1/r}$, and $M_0 = \exp(\ln x)$ is the geometric mean. $M_1 = \bar{x}$. The power mean increases with r , and all power means are equal only when all values are equal (the power-mean inequality).

S2 The Jensen gap

The power-mean Jensen gap of order α (Eq. 1) is

$$J_\alpha(x) = \frac{\langle x^\alpha \rangle}{\bar{x}^\alpha} - 1. \quad (\text{S4})$$

Jensen’s inequality says that for a convex function φ , $\langle \varphi(x) \rangle \geq \varphi(\bar{x})$, with equality only when all values are equal. For a concave function the inequality reverses. The function $\varphi(x) = x^\alpha$ is convex for $\alpha > 1$ and for $\alpha < 0$, and concave for $0 < \alpha < 1$. So $\langle x^\alpha \rangle - \bar{x}^\alpha$, and with it J_α , is positive for $\alpha > 1$ and $\alpha < 0$ and negative for $0 < \alpha < 1$, unless all values are equal. In both cases J_α has the sign of $\alpha(\alpha - 1)$.

Numerical check: sign violations (gap sign, divergence sign by order)

Since $\langle x^\alpha \rangle = M_\alpha^\alpha$ and $\bar{x} = M_1$, the gap is also

$$1 + J_\alpha = \left(\frac{M_\alpha}{M_1} \right)^\alpha. \quad (\text{S5})$$

At order 2, Eq. (S2) gives

$$J_2 = \frac{\langle x^2 \rangle}{\bar{x}^2} - 1 = \text{CV}_x^2. \quad (\text{S6})$$

S3 The identity, Eq. 2

Step 1, the power sum. Raise Eq. (S3) to the power α and sum over i :

$$\sum_i p_i^\alpha = \frac{\sum_i x_i^\alpha}{(K\bar{x})^\alpha} = \frac{K\langle x^\alpha \rangle}{K^\alpha \bar{x}^\alpha} = K^{1-\alpha} \frac{\langle x^\alpha \rangle}{\bar{x}^\alpha} = K^{1-\alpha} (1 + J_\alpha). \quad (\text{S7})$$

Step 2, the entropy. Take $(1 - \alpha)^{-1} \ln$ of Eq. (S7):

$$H_\alpha = \frac{1}{1 - \alpha} \left[(1 - \alpha) \ln K + \ln(1 + J_\alpha) \right] = \ln K + \frac{\ln(1 + J_\alpha)}{1 - \alpha}. \quad (\text{S8})$$

Step 3, the entropy deficit. Move $\ln K$ to the left:

$$\ln K - H_\alpha = \frac{\ln(1 + J_\alpha)}{\alpha - 1}. \quad (\text{S9})$$

Step 4, the divergence. The Rényi divergence of p from u is $\text{Div}_\alpha(p \| u) = (\alpha - 1)^{-1} \ln \sum_i p_i^\alpha u_i^{1-\alpha}$. Since $u_i^{1-\alpha} = K^{\alpha-1}$ for every i ,

$$\text{Div}_\alpha(p \| u) = \frac{1}{\alpha - 1} \left[(\alpha - 1) \ln K + \ln \sum_i p_i^\alpha \right] = \ln K - \frac{\ln \sum_i p_i^\alpha}{1 - \alpha} = \ln K - H_\alpha. \quad (\text{S10})$$

Step 5, the power-mean form. Insert Eq. (S5) into Eq. (S9):

$$\ln K - H_\alpha = \frac{1}{\alpha - 1} \ln \left(\frac{M_\alpha}{M_1} \right)^\alpha = \frac{\alpha}{\alpha - 1} \ln \frac{M_\alpha}{M_1}. \quad (\text{S11})$$

Steps 3 to 5 together are Eq. 2.

Step 6, the sign. For $\alpha > 1$, $M_\alpha \geq M_1$ and $\alpha/(\alpha - 1) > 0$, so the deficit is nonnegative. For $0 < \alpha < 1$, $M_\alpha \leq M_1$, so the logarithm is nonpositive, and $\alpha/(\alpha - 1) < 0$, so the deficit is again nonnegative. In both cases it is zero only when all values are equal.

Step 7, monotonicity. For fixed α , the right side of Eq. (S9) is a strictly monotone function of J_α , increasing for $\alpha > 1$ and decreasing for $\alpha < 1$. Knowing one of the entropy deficit and the Jensen gap of the same order therefore fixes the other.

Numerical check: Eq. 2 identity, every form, all orders | Eq. 2 derivation: $\sum p_i^\alpha = K^{1-\alpha}(1 + J_\alpha)$

S4 Order 1, the Shannon case

The limit. At $\alpha = 1$ both $\ln \sum_i p_i^\alpha$ and $1 - \alpha$ vanish. By l'Hôpital's rule, with $\frac{d}{d\alpha} \sum_i p_i^\alpha = \sum_i p_i^\alpha \ln p_i$ and $\sum_i p_i = 1$,

$$H_1 = \lim_{\alpha \rightarrow 1} \frac{\ln \sum_i p_i^\alpha}{1 - \alpha} = \frac{\sum_i p_i \ln p_i}{-1} = - \sum_i p_i \ln p_i. \quad (\text{S12})$$

The deficit is the Kullback–Leibler divergence. Since $\sum_i p_i = 1$,

$$\ln K - H_1 = \sum_i p_i \ln K + \sum_i p_i \ln p_i = \sum_i p_i \ln \frac{p_i}{1/K} = \text{Div}_{\text{KL}}(p \parallel u). \quad (\text{S13})$$

In terms of the values. By Eq. (S3), $Kp_i = x_i/\bar{x}$, so

$$\ln K - H_1 = \sum_i p_i \ln \frac{x_i}{\bar{x}} = \sum_i p_i \ln x_i - \ln \bar{x}. \quad (\text{S14})$$

This is Theil's index of the shares p_i .

The Jensen gap of $x \ln x$. For $\varphi(x) = x \ln x$, again using Eq. (S3),

$$\frac{\langle x \ln x \rangle - \bar{x} \ln \bar{x}}{\bar{x}} = \sum_i \frac{x_i}{K\bar{x}} \ln x_i - \ln \bar{x} = \sum_i p_i \ln x_i - \ln \bar{x}, \quad (\text{S15})$$

which is Eq. (S14).

The reverse direction. With the roles of p and u exchanged,

$$\text{Div}_{\text{KL}}(u \parallel p) = \sum_i \frac{1}{K} \ln \frac{1/K}{p_i} = - \left\langle \ln \frac{x}{\bar{x}} \right\rangle = \ln \bar{x} - \langle \ln x \rangle = \ln \frac{M_1}{M_0}. \quad (\text{S16})$$

It can also be written $\text{Div}_{\text{KL}}(u \parallel p) = H(u, p) - \ln K$, where $H(u, p) = - \sum_i u_i \ln p_i$ is the cross-entropy of u relative to p , not an entropy of p .

Numerical check: `uniform log gap ln(mean) - mean(ln) = KL(u||p) | Shannon deficit = Jensen gap of x ln x, scaled by mean`

S5 Other special orders

Order ∞ . As $\alpha \rightarrow \infty$, the largest proportion dominates $\sum_i p_i^\alpha$. If m proportions share the largest value $p_{\max}$, then $\sum_i p_i^\alpha = m p_{\max}^\alpha (1 + o(1))$, and

$$H_\infty = \lim_{\alpha \rightarrow \infty} \frac{\ln m + \alpha \ln p_{\max}}{1 - \alpha} = - \ln p_{\max}, \quad \ln K - H_\infty = \ln(K p_{\max}) = \ln \frac{x_{\max}}{\bar{x}}. \quad (\text{S17})$$

Numerical check: `order inf: ln K - H_inf = ln(max x / mean x)`

Order 2. Eq. (S7) with $\alpha = 2$ and Eq. (S6) give $\sum_i p_i^2 = (1 + \text{CV}_x^2)/K$. Then

$$\ln K - H_2 = \ln(1 + \text{CV}_x^2), \quad \text{PR} = \frac{1}{\sum_i p_i^2} = \frac{K}{1 + \text{CV}_x^2}, \quad \frac{\text{PR}}{K} = \frac{1}{1 + \text{CV}_x^2}. \quad (\text{S18})$$

The chi-squared divergence. Expanding the square and using $\sum_i p_i = 1$,

$$\sum_i \frac{(p_i - u_i)^2}{u_i} = K \sum_i \left(p_i^2 - \frac{2p_i}{K} + \frac{1}{K^2} \right) = K \sum_i p_i^2 - 2 + 1 = (1 + \text{CV}_x^2) - 1 = \text{CV}_x^2. \quad (\text{S19})$$

Numerical check: order 2: PR/K = 1/(1+CV^2) | order 2: CV^2 = chi^2(p||u)

S6 The geometric form, Eq. 3

Let θ be the angle between the vector $x = (x_1, \dots, x_K)$ and the isotropic direction $\mathbf{1} = (1, \dots, 1)$. Then

$$\cos^2 \theta = \frac{(x \cdot \mathbf{1})^2}{|x|^2 |\mathbf{1}|^2} = \frac{(\sum_i x_i)^2}{K \sum_i x_i^2} = \frac{K^2 \bar{x}^2}{K \cdot K \langle x^2 \rangle} = \frac{\bar{x}^2}{\langle x^2 \rangle} = \frac{1}{1 + \text{CV}_x^2} = \frac{\text{PR}}{K}, \quad (\text{S20})$$

using Eqs. (S2) and (S18). Then $\tan^2 \theta = 1/\cos^2 \theta - 1 = \text{CV}_x^2$, and since the values are positive, $0 \leq \theta < \pi/2$ and $\tan \theta = \text{CV}_x$. Also $\sin^2 \theta = 1 - \cos^2 \theta = \text{CV}_x^2/(1 + \text{CV}_x^2)$.

Numerical check: PR/K = cos^2 theta | CV = tan theta

S7 Near isotropy, Eq. 4

Write $x_i = \bar{x}(1 + \varepsilon_i)$. Averaging gives $\langle \varepsilon \rangle = 0$, and $\langle \varepsilon^2 \rangle = \sigma_x^2/\bar{x}^2 = \text{CV}_x^2$. Expanding $(1 + \varepsilon_i)^\alpha$ to second order,

$$1 + J_\alpha = \langle (1 + \varepsilon)^\alpha \rangle = 1 + \alpha \langle \varepsilon \rangle + \frac{1}{2} \alpha(\alpha - 1) \langle \varepsilon^2 \rangle + O(\varepsilon^3) = 1 + \frac{1}{2} \alpha(\alpha - 1) \text{CV}_x^2 + O(\varepsilon^3). \quad (\text{S21})$$

Since $\ln(1 + J) = J + O(J^2)$, Eq. (S9) gives

$$\ln K - H_\alpha = \frac{J_\alpha}{\alpha - 1} + O(\varepsilon^3) = \frac{\alpha}{2} \text{CV}_x^2 + O(\varepsilon^3). \quad (\text{S22})$$

Carrying the expansion of $(1 + \varepsilon_i)^\alpha$ one term further adds $\frac{1}{6} \alpha(\alpha - 1)(\alpha - 2) \langle \varepsilon^3 \rangle$ to $1 + J_\alpha$. Since J_α^2 is $O(\varepsilon^4)$,

$$\ln K - H_\alpha = \frac{\alpha}{2} \text{CV}_x^2 + \frac{\alpha(\alpha - 2)}{6} \langle \varepsilon^3 \rangle + O(\varepsilon^4). \quad (\text{S23})$$

The cubic coefficient grows without bound with α , so the approximation is not uniform in the order. At $\alpha = \infty$ the deficit $\ln(x_{\max}/\bar{x}) = \ln(1 + \varepsilon_{\max}) \approx \varepsilon_{\max}$ is first order.

Numerical check: Eq. 4: J_alpha / (alpha(alpha-1)/2 CV^2) | Eq. 4: (ln K - H.alpha) / (alpha/2 CV^2) | Supplement S7: cubic term | Eq. 4: at alpha = infinity the deficit is first order

S8 Divergences are not metrics

A metric must be symmetric and satisfy the triangle inequality. At order $\frac{1}{2}$, $\text{Div}_{1/2}(p \| q) = -2 \ln \sum_i \sqrt{p_i q_i}$, which is symmetric in p and q . It fails the triangle inequality. For two outcomes, take $p = (0.9, 0.1)$, $q = (0.5, 0.5)$ and $r = (0.1, 0.9)$. Then $\text{Div}_{1/2}(p \| q) = \text{Div}_{1/2}(q \| r) = 0.2231$, but $\text{Div}_{1/2}(p \| r) = -2 \ln 0.6 = 1.0217$, which exceeds their sum. Other positive orders are not symmetric. For example, at order 1, $\text{Div}_{\text{KL}}(p \| u) \neq \text{Div}_{\text{KL}}(u \| p)$ in general, as Section S4 shows.

Numerical check: order 1/2 is symmetric | orders 1, 2 and 4 are asymmetric | order 1/2 fails the triangle inequality | Supplement: triangle example

S9 The diffusion tensor

Take $x = \lambda$, the eigenvalues of the tensor, with $K = 3$, mean $\bar{\lambda}$ and normalized eigenvalues $p_i^{(\lambda)} = \lambda_i / \sum_j \lambda_j$.

FA, Eq. 5. Fractional anisotropy is $FA^2 = \frac{3}{2} \sum_i (\lambda_i - \bar{\lambda})^2 / \sum_i \lambda_i^2$. With divisor 3, $\sum_i (\lambda_i - \bar{\lambda})^2 = 3\sigma_\lambda^2 = 3\bar{\lambda}^2 CV_\lambda^2$ and, by Eq. (S2), $\sum_i \lambda_i^2 = 3\langle \lambda^2 \rangle = 3\bar{\lambda}^2(1 + CV_\lambda^2)$. So

$$FA^2 = \frac{3}{2} \frac{CV_\lambda^2}{1 + CV_\lambda^2}. \quad (S24)$$

By Eq. (S18), $Div_2 = \ln 3 - H_2 = \ln(1 + CV_\lambda^2)$, so $e^{-Div_2} = 1/(1 + CV_\lambda^2)$ and $1 - e^{-Div_2} = CV_\lambda^2/(1 + CV_\lambda^2)$. Hence

$$FA^2 = \frac{3}{2} (1 - e^{-Div_2}) = \frac{3}{2} (1 - PR_\lambda/3) = \frac{3}{2} \sin^2 \theta, \quad (S25)$$

since $PR_\lambda/3 = e^{-Div_2}$ (Eq. (S18)) and $\sin^2 \theta = CV_\lambda^2/(1 + CV_\lambda^2)$.

FA and the order-2 entropy. From $FA^2 = \frac{3}{2}(1 - e^{H_2}/3)$, solve for H_2 : $e^{H_2} = 3 - 2FA^2$, so

$$H_2 = \ln(3 - 2FA^2), \quad \ln 3 - H_2 = -\ln(1 - \frac{2}{3}FA^2). \quad (S26)$$

Near isotropy the second form is $\frac{2}{3}FA^2$ to leading order.

Numerical check: tensor orders, FA, QA and VR relations | $\ln 3 - H_2(\text{lambda}) = -\ln(1 - 2FA^2/3)$

The order-1 member near isotropy. Near isotropy $FA^2 = \frac{3}{2} CV_\lambda^2 / (1 + CV_\lambda^2) \approx \frac{3}{2} CV_\lambda^2$, so $CV_\lambda^2 \approx \frac{2}{3} FA^2$. By Eq. 4 at $\alpha = 1$ the order-1 member is then $\approx \frac{1}{2} CV_\lambda^2 = FA^2/3$ to leading order.

Numerical check: order-1 member / (FA^2/3) near isotropy (FA < 0.05), median

The trace form of Özarslan et al. The trace-normalized tensor $R = \mathbf{D}/\text{tr}\mathbf{D}$ has eigenvalues $p_i^{(\lambda)}$, so $\text{tr}R^2 = \sum_i (p_i^{(\lambda)})^2 = e^{-H_2}$. Then $FA^2 = \frac{3}{2} (1 - 1/(3\text{tr}R^2))$, which is $FA = \sqrt{(3 - 1/\text{tr}R^2)/2}$.

Numerical check: Ozarslan Eq. 11, FA from trace(R^2), and trace(R^2) = exp(-H_2)

Procrustes anisotropy. Dryden et al. define PA as FA computed on $q_i = \sqrt{\lambda_i}$. Since $\sum_i (q_i - \bar{q})^2 = \sum_i q_i^2 - 3\bar{q}^2$ and $\sum_i q_i^2 = \sum_i \lambda_i$,

$$PA^2 = \frac{3}{2} \left(1 - \frac{(\sum_i q_i)^2}{3 \sum_i q_i^2} \right). \quad (S27)$$

At order $\frac{1}{2}$, with $u_i^{1/2} = 3^{-1/2}$, $Div_{1/2} = -2 \ln \sum_i \sqrt{p_i^{(\lambda)}/3} = \ln 3 - 2 \ln \sum_i \sqrt{p_i^{(\lambda)}}$, so $e^{-Div_{1/2}} = (\sum_i \sqrt{p_i^{(\lambda)}})^2 / 3$. Since $\sqrt{p_i^{(\lambda)}} = q_i / \sqrt{\sum_j \lambda_j}$, $(\sum_i \sqrt{p_i^{(\lambda)}})^2 = (\sum_i q_i)^2 / \sum_i \lambda_i$. Therefore $PA^2 = \frac{3}{2} (1 - e^{-Div_{1/2}})$.

Numerical check: Procrustes anisotropy (Dryden 2009): $PA^2 = (3/2)(1 - \exp(-Div_{1/2}))$

Relative anisotropy. In one common convention $RA = \sqrt{\sum_i (\lambda_i - \bar{\lambda})^2} / (\sqrt{3} \bar{\lambda}) = \sigma_\lambda / \bar{\lambda} = CV_\lambda$. Other conventions multiply this by a constant.

Quantitative anisotropy, Eq. 6. By Eq. (S14) with $x = \lambda$, $\ln 3 - H_1(p^{(\lambda)}) = \sum_i p_i^{(\lambda)} \ln \lambda_i - \ln \bar{\lambda}$. The von Neumann entropy of R is $-\text{tr}(R \ln R) = -\sum_i p_i^{(\lambda)} \ln p_i^{(\lambda)} = H_1(p^{(\lambda)})$, because R has eigenvalues $p_i^{(\lambda)}$. So the order-1 member is $\ln 3$ minus the von Neumann entropy of R .

The volume ratio. $\text{VR} = \lambda_1 \lambda_2 \lambda_3 / \bar{\lambda}^3$. The geometric mean is $M_0 = (\lambda_1 \lambda_2 \lambda_3)^{1/3}$, so $\text{VR} = (M_0/M_1)^3$, and by Eq. (S16)

$$-\frac{1}{3} \ln \text{VR} = \ln \frac{M_1}{M_0} = \text{Div}_{\text{KL}}(u \parallel p^{(\lambda)}). \quad (\text{S28})$$

FA and tensor mode fix the normalized eigenvalues. The normalized eigenvalues sum to 1, so they have two degrees of freedom. Write $d = p^{(\lambda)} - u$. Its components sum to zero, so d lies in a plane, and d is orthogonal to u . Hence $\sum_i (p_i^{(\lambda)})^2 = |u|^2 + |d|^2 = \frac{1}{3} + |d|^2$, and $|d|$ fixes $\sum_i (p_i^{(\lambda)})^2$ and therefore FA. In the plane, use the orthonormal basis $e_1 = (2, -1, -1)/\sqrt{6}$, $e_2 = (0, 1, -1)/\sqrt{2}$ and write $d = |d|(\cos \phi e_1 + \sin \phi e_2)$. The product of the components of d is $|d|^3 \cos 3\phi / (3\sqrt{6})$, so the tensor mode, $3\sqrt{6} \prod_i d_i / |d|^3$, equals $\cos 3\phi$. With $\lambda_1 \geq \lambda_2 \geq \lambda_3$, ϕ runs from 0 (linear, mode +1) to $\pi/3$ (planar, mode -1), and $\cos 3\phi$ is one-to-one on that range. So FA fixes $|d|$, mode fixes ϕ , and together they fix $p^{(\lambda)}$ and every entropy of it.

Numerical check: Supplement: mode = cos 3 phi and sum p^2 = 1/3 + |d|^2

The order-1 band at fixed FA. Scale the eigenvalues so that $\lambda_1 = 1$, and let $0 \leq t \leq 1$. Linear tensors are $(1, t, t)$, planar tensors $(1, 1, t)$, and tensors with a zero eigenvalue $(1, t, 0)$. By the definition of FA in Eq. 5, the three families have

$$\text{FA}^2 = \frac{(1-t)^2}{1+2t^2}, \quad \text{FA}^2 = \frac{(1-t)^2}{2+t^2}, \quad \text{FA}^2 = \frac{t^2-t+1}{1+t^2}. \quad (\text{S29})$$

The planar family reaches its largest FA, $1/\sqrt{2}$, at $t = 0$, the flat disc $(1, 1, 0)$. The zero-eigenvalue family runs from the same disc at $t = 1$ to the stick $(1, 0, 0)$, with $\text{FA} = 1$, at $t = 0$. A fixed FA fixes $\sum_i (p_i^{(\lambda)})^2$, the index of coincidence. Harremoës and Topsøe derived the joint range of the Shannon entropy and the index of coincidence (cited in the paper). For three outcomes, its upper boundary is the linear family. Its lower boundary is the planar family up to $\text{FA} = 1/\sqrt{2}$ and the zero-eigenvalue family above it. We do not repeat their proof.

Numerical check: Supplement S9: FA of the linear, planar and zero-eigenvalue families | random eigenvalue triples inside the linear/planar/zero-eigenvalue interval | interval width at FA 0.2 | interval width at FA 0.5

S10 The directional profile

GFA, Eq. 7. For N values D_i , Tuch's generalized fractional anisotropy is the standard deviation, with divisor $N - 1$, over the root mean square:

$$\text{GFA}^2 = \frac{N \sum_i (D_i - \bar{D})^2}{(N-1) \sum_i D_i^2} = \frac{N}{N-1} \frac{N \sigma_D^2}{N \langle D^2 \rangle} = \frac{N}{N-1} \frac{\text{CV}_D^2}{1 + \text{CV}_D^2}. \quad (\text{S30})$$

By Eq. (S18) and the geometric form this equals $\frac{N}{N-1} (1 - \text{PR}/N) = \frac{N}{N-1} \sin^2 \theta$. With $N = 3$ the prefactor is $\frac{3}{2}$ and this is Eq. 5.

Numerical check: Eq. 7: $\text{GFA}^2 \text{ (Tuch)} = N/(N-1) \text{ CV}^2/(1+\text{CV}^2) = N/(N-1) (1 - \text{PR}/N) = N/(N-1) \sin^2 \theta$

The CVD of Aja-Fernández et al. With $\log E_i = -bD_i$, their sample variance and second moment are

$$\mathcal{V} = \frac{N}{N-1} [\langle (\log E)^2 \rangle - \langle \log E \rangle^2] = \frac{N}{N-1} b^2 \sigma_D^2, \quad \langle (\log E)^2 \rangle = b^2 \langle D^2 \rangle. \quad (\text{S31})$$

So $\text{CVD}^2 = \mathcal{V} / \langle (\log E)^2 \rangle = \frac{N}{N-1} \sigma_D^2 / \langle D^2 \rangle = \text{GFA}^2$.

Numerical check: CVD^2 of Aja-Fernandez 2018 = GFA^2 of the profile, Eq. (7)

The signal domain. The signals $S_i/S_0 = e^{-bD_i}$ are positive, so the identity applies to them as well. Near isotropy write $D_i = \bar{D} + \delta_i$ with $\langle \delta \rangle = 0$ and $\langle \delta^2 \rangle = \sigma_D^2$. Then $e^{-bD_i} = e^{-b\bar{D}} e^{-b\delta_i} \approx e^{-b\bar{D}} (1 - b\delta_i)$. The signals have mean $\approx e^{-b\bar{D}}$ and standard deviation $\approx e^{-b\bar{D}} b\sigma_D$, so their coefficient of variation is $\approx b\sigma_D$. By Section S7 the entropy deficits of the two domains are $\approx \frac{\alpha}{2} \text{CV}_D^2$ and $\approx \frac{\alpha}{2} (b\sigma_D)^2$, and their ratio is

$$\frac{(b\sigma_D)^2}{(\sigma_D/\bar{D})^2} = (b\bar{D})^2. \quad (\text{S32})$$

Numerical check: Sec 4.1 near isotropy: signal/diffusivity deficit ratio vs $(b\bar{D})^2$

S11 The link to the tensor, Eq. 8

For a single Gaussian compartment the profile is $D(\hat{g}) = \hat{g}^\top \mathbf{D} \hat{g}$. Averages over the unit sphere of products of the components of $\hat{g}$ are

$$\langle g_i g_j \rangle = \frac{1}{3} \delta_{ij}, \quad \langle g_i g_j g_k g_l \rangle = \frac{1}{15} (\delta_{ij} \delta_{kl} + \delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}). \quad (\text{S33})$$

The mean of the profile is $\langle D \rangle = \sum_{ij} D_{ij} \langle g_i g_j \rangle = \text{tr} \mathbf{D} / 3 = \bar{\lambda}$. Its mean square is

$$\langle D^2 \rangle = \sum_{ijkl} D_{ij} D_{kl} \langle g_i g_j g_k g_l \rangle = \frac{1}{15} [(\text{tr} \mathbf{D})^2 + 2 \text{tr} \mathbf{D}^2] = \frac{1}{15} [9\bar{\lambda}^2 + 2 \sum_i \lambda_i^2]. \quad (\text{S34})$$

Its variance is

$$\langle D^2 \rangle - \langle D \rangle^2 = \frac{1}{15} [2 \sum_i \lambda_i^2 - 6\bar{\lambda}^2] = \frac{2}{15} \sum_i (\lambda_i - \bar{\lambda})^2 = \frac{2}{5} \sigma_\lambda^2, \quad (\text{S35})$$

using $\sum_i (\lambda_i - \bar{\lambda})^2 = \sum_i \lambda_i^2 - 3\bar{\lambda}^2 = 3\sigma_\lambda^2$. Dividing by $\bar{\lambda}^2$,

$$\text{CV}_D^2 = \frac{2}{5} \text{CV}_\lambda^2. \quad (\text{S36})$$

Equal-weight averages over N near-uniform directions approximate these sphere averages.

The linear limit. For $\lambda \propto (1, 0, 0)$, $\text{CV}_\lambda^2 = 2$ and $\text{CV}_D^2 = \frac{4}{5}$. The normalized second moments are $1 + \text{CV}_\lambda^2 = 3$ and $1 + \text{CV}_D^2 = \frac{9}{5}$, whose ratio is $\frac{3}{5}$, the value Özarslan et al. report for the profile against the eigenvalues.

Numerical check: Sec 4.2 sphere moments | $\text{CV}_D^2 / \text{CV}_\lambda^2$ on a dense sphere, 50 random tensors | linear tensor: profile's normalized second moment / eigenvalues' = 3/5

S12 The order-1 link between profile and tensor

In the eigenvector basis, $D(\hat{g}) = \sum_i w_i(\hat{g}) \lambda_i$ with $w_i(\hat{g}) = (\hat{g} \cdot e_i)^2 \geq 0$ and $\sum_i w_i = 1$. So each direction's diffusivity is a weighted mean of the eigenvalues. Let $f(x) = x \ln x$, which is convex. Jensen's inequality for each direction gives $f(D(\hat{g})) \leq \sum_i w_i(\hat{g}) f(\lambda_i)$. Averaging over the sphere, with $\langle w_i \rangle = \frac{1}{3}$,

$$\langle f(D) \rangle \leq \frac{1}{3} \sum_i f(\lambda_i) = \langle f(\lambda) \rangle. \quad (\text{S37})$$

The profile and the eigenvalues have the same mean $\bar{\lambda}$, so subtracting $f(\bar{\lambda})$ and dividing by $\bar{\lambda}$ turns both sides into order-1 deficits (Section S4). The profile's order-1 deficit is therefore at most the

eigenvalues', with equality only at isotropy. Near isotropy both deficits are $\approx \frac{1}{2}CV^2$, so their ratio tends to $CV_D^2/CV_\lambda^2 = \frac{2}{5}$.

Numerical check: order-1 link: Jensen violations over the eigenvalue simplex | order-1 link: median ratio near isotropy (FA < 0.05)

S13 The chain rule at order 1

Split the K values into groups g . Let $P_g = \sum_{i \in g} p_i$ be a group's proportion of the total and $U_g = |g|/K$ its fraction of the values. Within a group, $p_{i|g} = p_i/P_g$ and $u_{i|g} = 1/|g|$, so $p_i = P_g p_{i|g}$ and $u_i = U_g u_{i|g}$. Then

$$\text{Div}_{\text{KL}}(p \parallel u) = \sum_g \sum_{i \in g} P_g p_{i|g} \left[\ln \frac{P_g}{U_g} + \ln \frac{p_{i|g}}{u_{i|g}} \right] = \text{Div}_{\text{KL}}(P \parallel U) + \sum_g P_g \text{Div}_{\text{KL}}(p_{\cdot|g} \parallel u_{\cdot|g}), \quad (\text{S38})$$

using $\sum_{i \in g} p_{i|g} = 1$. The order-1 deficit is the proportion-weighted mean of the within-group deficits plus the divergence of the groups' proportions from their fractions of the values.

Numerical check: order-1 deficit = share-weighted within-group deficits + between-group divergence